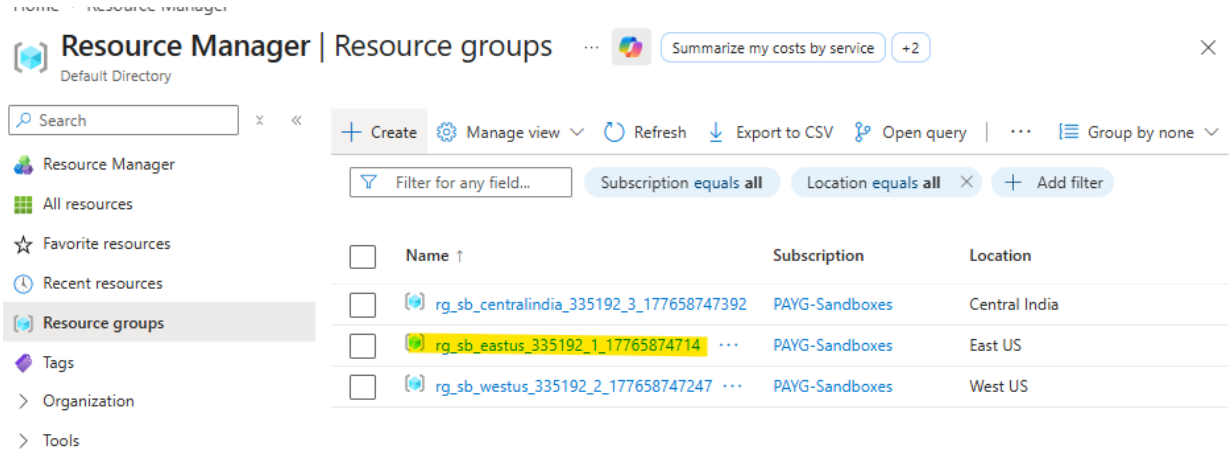
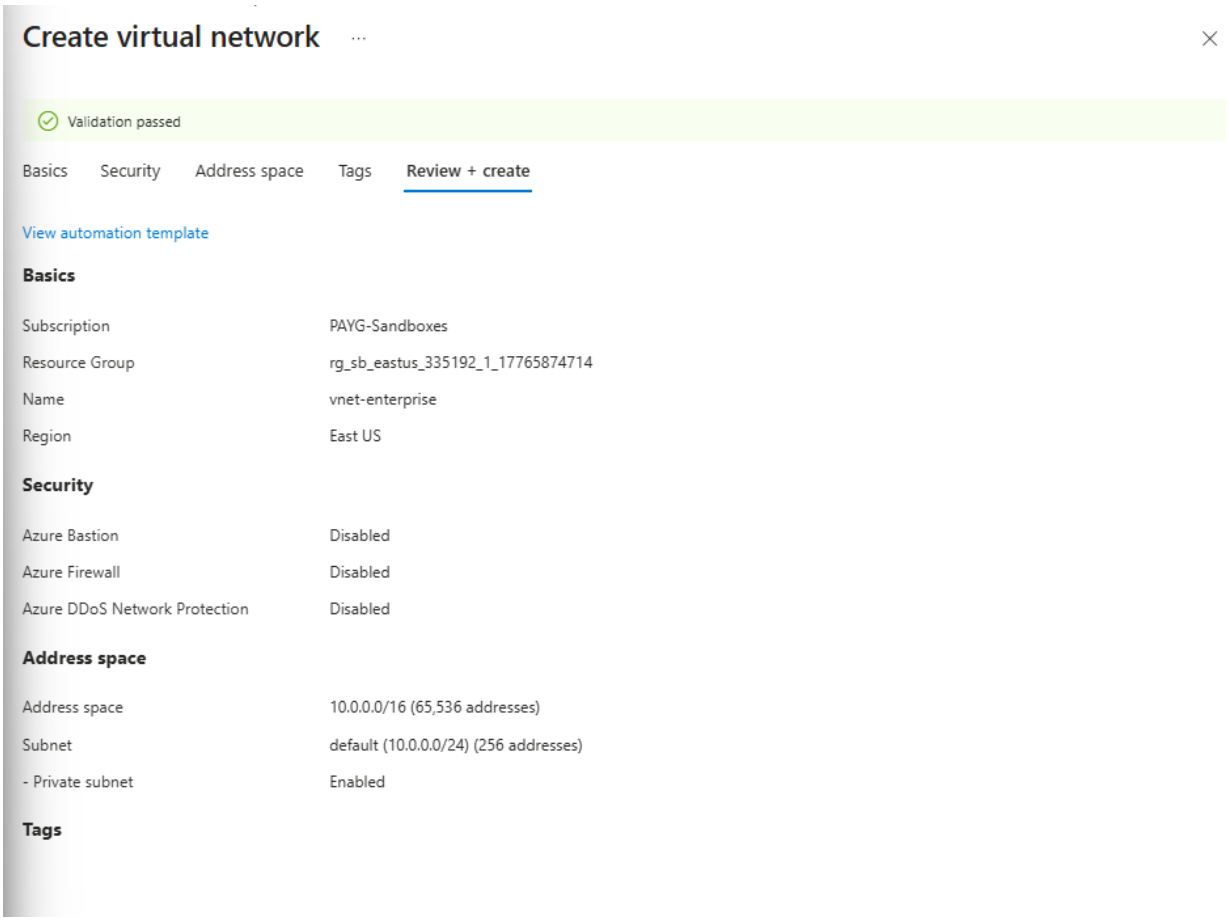
1. Resource Organization

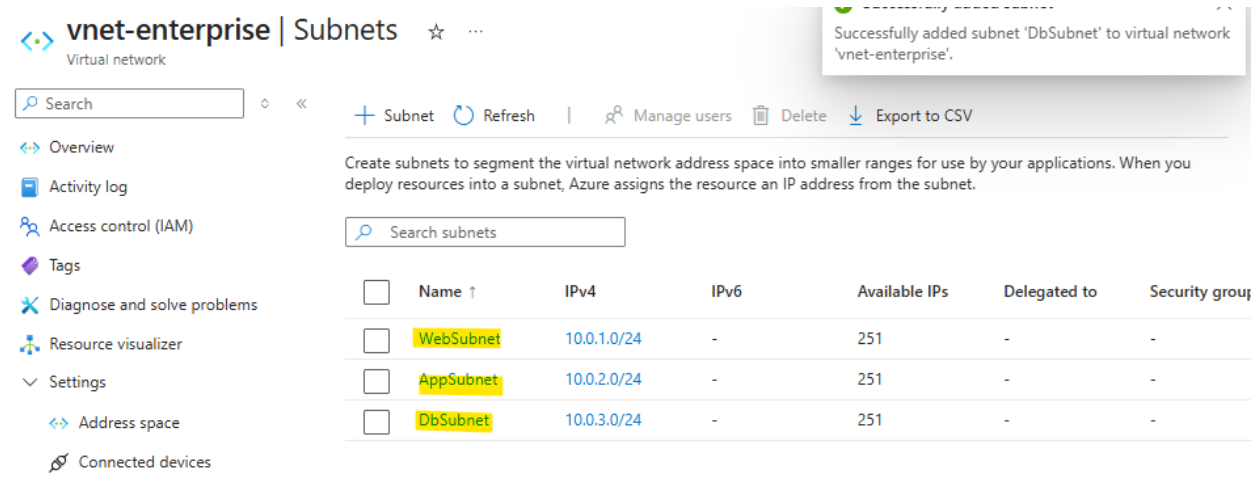


2. Networking Layer

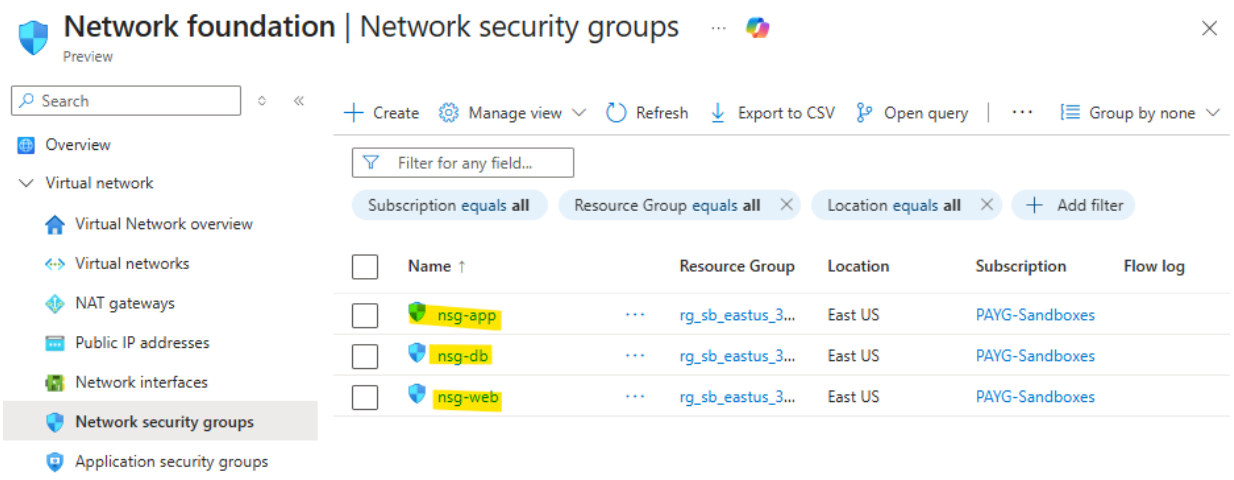
Create a Virtual Network (VNet)



Divide the network into Subnets



Apply Network Security Groups (NSGs) with proper rules



nsg-web | Inbound security rules

Network security group

Successfully created security rule 'allow-ssh'.

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties

Add | Hide default rules | Refresh | Delete | Give feedback

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority.
Learn more

Filter by name

Port == all | Protocol == all | Source == all | Destination == all | Action == all

Priority	Name	Port	Protocol	Source
100	allow-http	80	TCP	Any
110	allow-ssh	22	TCP	46.40.244.168
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalan···	Any	Any	AzureLoadBalan
65500	DenyAllInBound	Any	Any	Any

nsg-app | Inbound security rules

Network security group

Created security rule
Successfully created security rule 'allow-web-to-app'.

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Inbound security rules
Outbound security rules
Network interfaces
Subnets
Properties

Add | Hide default rules | Refresh | Delete | Give feedback

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority.
Learn more

Filter by name

Port == all | Protocol == all | Source == all | Destination == all | Action == all

Priority	Name	Port	Protocol	Source
100	allow-web-to-app	8080	Any	Any
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalan···	Any	Any	AzureLoadBalan
65500	DenyAllInBound	Any	Any	Any

nsg-db | Inbound security rules

Network security group

Successfully saved security rule 'allow-app-to-db'.

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
  - Inbound security rules
  - Outbound security rules
  - Network interfaces
  - Subnets
  - Properties

Add | Hide default rules | Refresh | Delete | Give feedback

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol to allow or deny the traffic. A security rule can't have the same priority and direction as an existing rule. You can't delete default security rules, but you can override them with rules that have a higher priority.
Learn more

Filter by name

Port == all | Protocol == all | Source == all | Destination == all | Action == all

Priority	Name	Port	Protocol	Source
100	allow-app-to-db	1433	Any	46.40.244.168
65000	AllowVnetInBound	Any	Any	VirtualNetwork
65001	AllowAzureLoadBalan···	Any	Any	AzureLoadBalan
65500	DenyAllInBound	Any	Any	Any

vnet-enterprise | Subnets

Virtual network

Subnet | Refresh | Manage users | Delete | Export to CSV

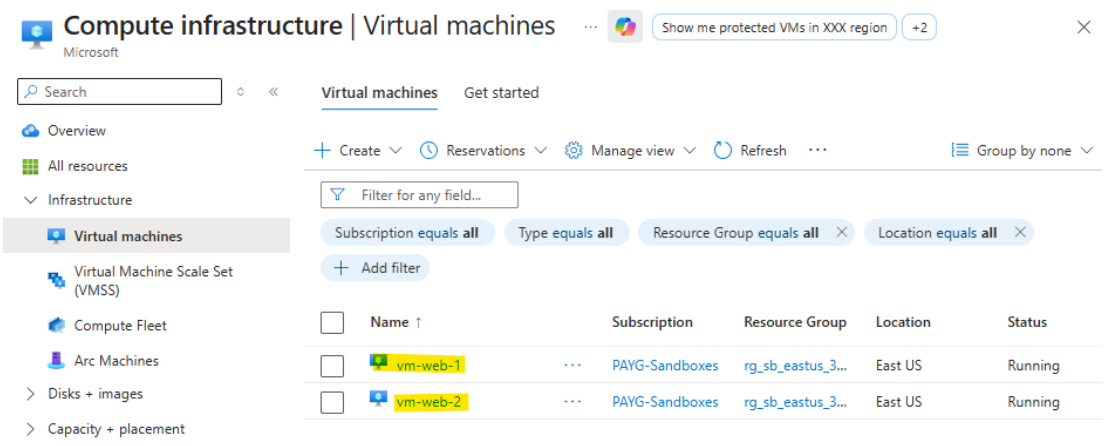
Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table
WebSubnet	10.0.1.0/24	-	251	-	nsg-web	-
DbSubnet	10.0.3.0/24	-	251	-	nsg-db	-
AppSubnet	10.0.2.0/24	-	251	-	nsg-app	-

3. Compute Layer

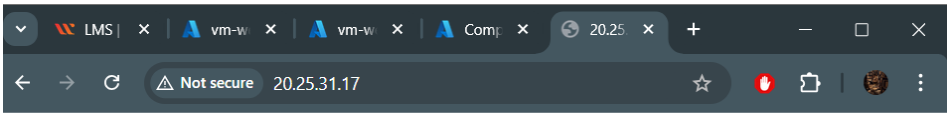
Deploy at least 2 Virtual Machines inside the Web Subnet



VM1

```
azureuser@vm-web-1:~$ sudo apt update
sudo apt install nginx -y
echo "Hello from VM1 created by Maryam Alshehhi" | sudo tee /var/www/html/index.html
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
```

```
azureuser@vm-web-1: ~
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (16.3 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 69212 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
 * Upgrading binary nginx                                                    [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Hello from VM1 created by Maryam Alshehhi
azureuser@vm-web-1:~$
```

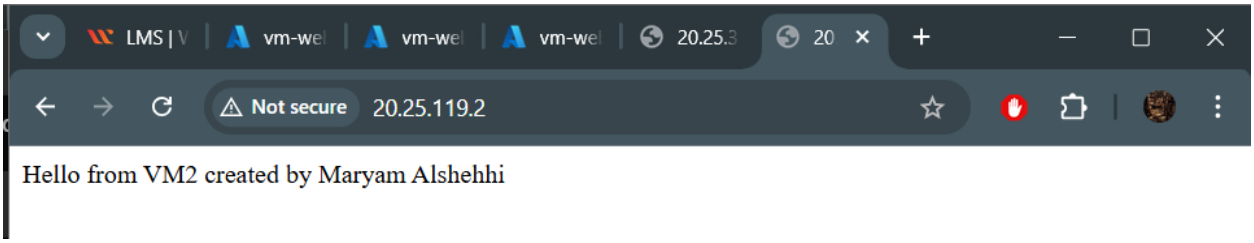


Hello from VM1 created by Maryam Alshehhi

VM2

```
azureuser@vm-web-2:~$ sudo apt update
sudo apt install nginx -y
echo "Hello from VM2 created by Maryam Alshehhi" | sudo tee /var/www/html/index.html
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
```

```
azureuser@vm-web-2: ~

Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (4151 kB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 69212 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
 * Upgrading binary nginx                                                                    [ OK ]
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
Hello from VM2 created by Maryam Alshehhi
azureuser@vm-web-2:~$
```



4. Load Balancing

Home › Load balancing and content delivery | Load balancers

Create load balancer

Basics | Frontend IP configuration | Backend pools | Inbound rules | Outbound rules | Tags | Review + create

Azure load balancer is a layer 4 load balancer that distributes incoming traffic among healthy virtual machine instances. Load balancers uses a hash-based distribution algorithm. By default, it uses a 5-tuple (source IP, source port, destination IP, destination port, protocol type) hash to map traffic to available servers. Load balancers can either be internet-facing where it is accessible via public IP addresses, or internal where it is only accessible from a virtual network. Azure load balancers also support Network Address Translation (NAT) to route traffic between public and private IP addresses. Learn more.

Project details

Subscription * : PAYG-Sandboxes

Resource group * : rg_sb_eastus_335192_1_17765874714

Create new

Instance details

Name * : lb-web

Region * : East US

SKU * : Standard (Distribute traffic to backend resources) / Gateway (Direct traffic to network virtual appliances)

Type * : Public / Internal

Tier * : Regional / Global

Home › Load balancing and content delivery | Load balancers › Create load balancer

Add backend pool

Name * : backend-web

Virtual network : vnet-enterprise (rg_sb_eastus_335192_1_17765874714)

The dropdown only shows virtual networks in the same subscription and location as the load balancer. If you don't see the one you're looking for, it's either in another subscription or location or you don't have access to it.

Backend Pool Configuration : NIC / IP address

IP configurations

IP configurations associated to virtual machines and virtual machine scale sets must be in same location as the load balancer and be in the same virtual network.

+ Add | Remove

Resource Name	Resource group	Type	IP confi...	IP Addr...	Availabi...
vm-web-1	rg_sb_eastus_335192_1_17765874714	Virtual machine	ipconfig1	10.0.1.6	-
vm-web-2	rg_sb_eastus_335192_1_17765874714	Virtual machine	ipconfig1	10.0.1.7	-

Home › Load balancing and content delivery | Load balancers

Create load balancer

Basics | Frontend IP configuration | Backend pools | Inbound rules | Outbound rules | Tags | Review + create

Load balancing rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. The load balancing rule uses a health probe to determine which backend instances are eligible to receive traffic.

+ Add a load balancing rule

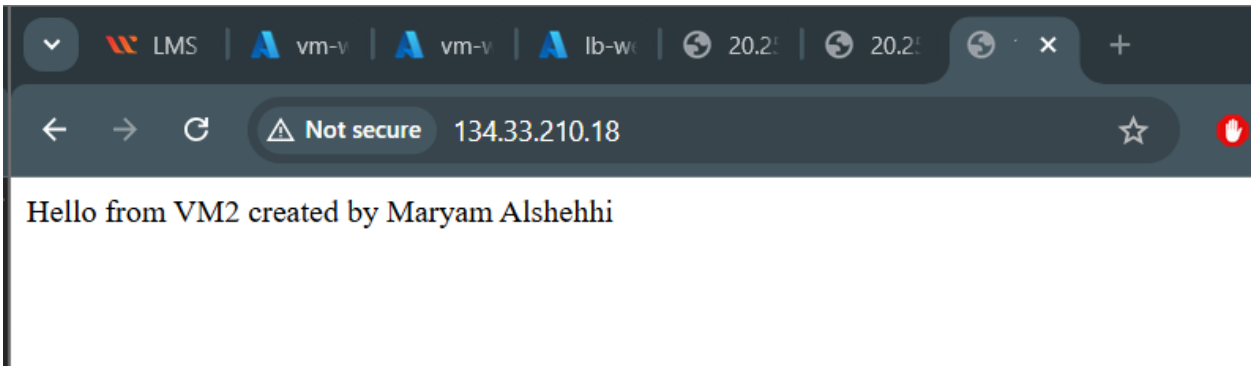
Name	Frontend IP config...	Backend pool	Health probe	Frontend Port	Backend port
http-rule	frontend-ip	backend-web	http-probe	80	80

Inbound NAT rule

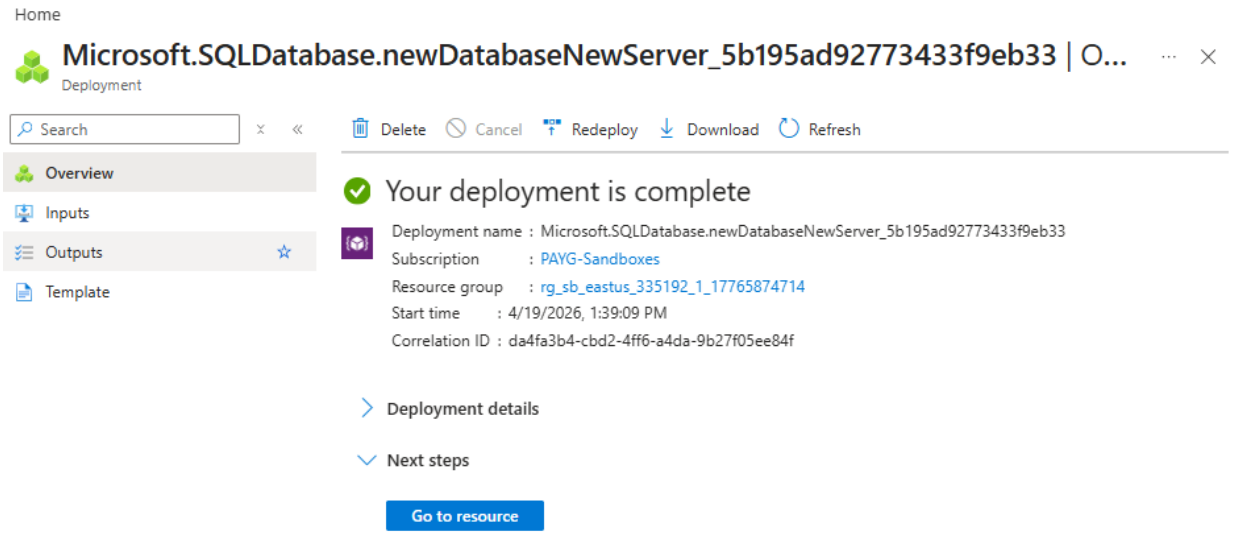
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

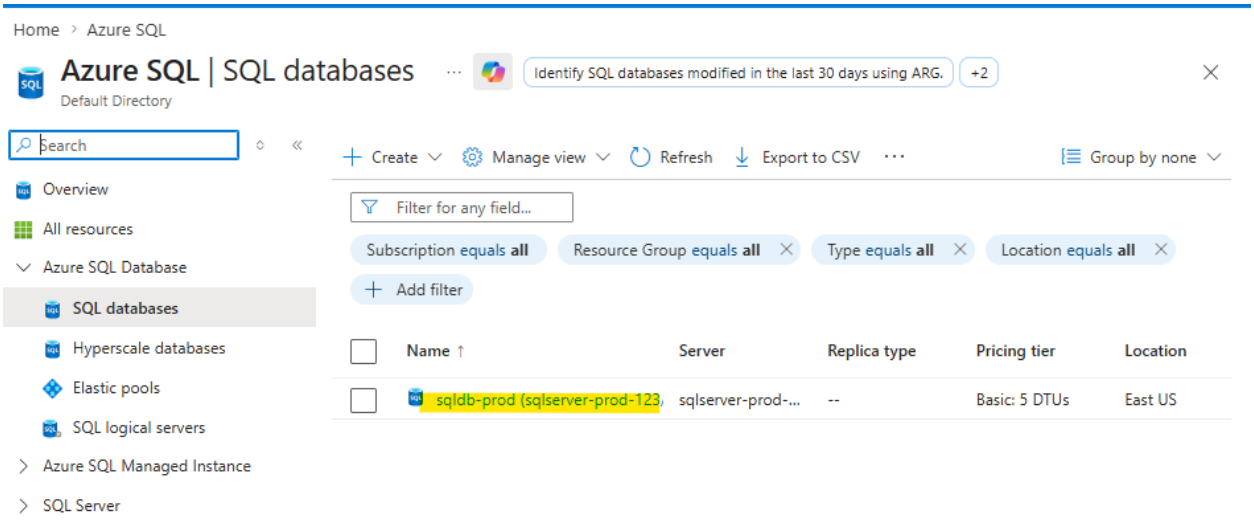
+ Add an inbound nat rule

Name	Frontend IP configuration	Service	Target	Frontend Port
Add a rule to get started				



5. Data Layer





(Disable Public Access + Allow access only from within the Virtual Network):

I CANNOT fully implement this in sandbox

6. Security

Home > Key vaults

Create a key vault

Basics | Access configuration | Networking | Tags | Review + create

Azure Key Vault is a cloud service used to manage keys, secrets, and certificates. Key Vault eliminates the need for developers to store security information in their code. It allows you to centralize the storage of your application secrets which greatly reduces the chances that secrets may be leaked. Key Vault also allows you to securely store secrets and keys backed by Hardware Security Modules or HSMs. The HSMs used are Federal Information Processing Standards (FIPS) 140-2 Level 2 validated. In addition, key vault provides logs of all access and usage attempts of your secrets so you have a complete audit trail for compliance.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : PAYG-Sandboxes

Resource group * : rg_sb_eastus_335192_1_17765874714

Create new

Instance details

Key vault name * : kv-enterprise1104

Region * : East US

Pricing tier * : Standard

Recovery options

Soft delete protection will automatically be enabled on this key vault. This feature allows you to recover or permanently delete a key vault and secrets for the duration of the retention period. This protection applies to the key vault and the secrets stored within the key vault.

To enforce a mandatory retention period and prevent the permanent deletion of key vaults or secrets prior to the retention period elapsing, you can turn on purge protection. When purge protection is enabled, secrets cannot be purged by users or by Microsoft.

Soft-delete : Enabled

Days to retain deleted vaults * : 90

Purge protection : Disable purge protection (allow key vault and objects to be purged during retention period)

Home › Key vaults

Create a key vault

Basics | Access configuration | Networking | Tags | Review + create

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines which operations the caller can execute. Learn more

Permission model

Grant data plane access by using a Azure RBAC or Key Vault access policy

- (●) Azure role-based access control (recommended)
- ( ) Vault access policy

Resource access

- [ ] Azure Virtual Machines for deployment
- [ ] Azure Resource Manager for template deployment
- [ ] Azure Disk Encryption for volume encryption

Home › Key vaults

Create a key vault

Basics | Access configuration | Networking | Tags | Review + create

You can connect to this key vault either publicly, via public IP addresses or service endpoints, or privately, using a private endpoint.

Enable public access [✓]

Public Access

Allow access from:

- (●) All networks
- ( ) Selected networks

Traffic from all public networks can access this resource. This is not recommended for private applications or environments. Learn more about key vault network security configuration

Private endpoint

Create a private endpoint to allow a private connection to this resource. Additional private endpoint connections can be created within the key vault or private link center. Learn more

+ Create a private endpoint

Name	Subscription	Resource group	Region	Subne
Click on add button to add private endpoint				

Home

kv-enterprise1104 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete Cancel Redeploy Download Refresh

Your deployment is complete

Deployment name : kv-enterprise1104
Subscription : PAYG-Sandboxes
Resource group : rg_sb_eastus_335192_1_17765874714
Start time : 4/19/2026, 1:50:47 PM
Correlation ID : d32848e3-8486-4382-a919-2d8b19fd8927

Deployment details

Next steps

Go to resource

Home

Key vaults

Default Directory (instructorwhizlabs.onmicrosoft.com)

How many Key Vaults do I have in total? | Identify non-compliant Key Vaults in my environment. | +1

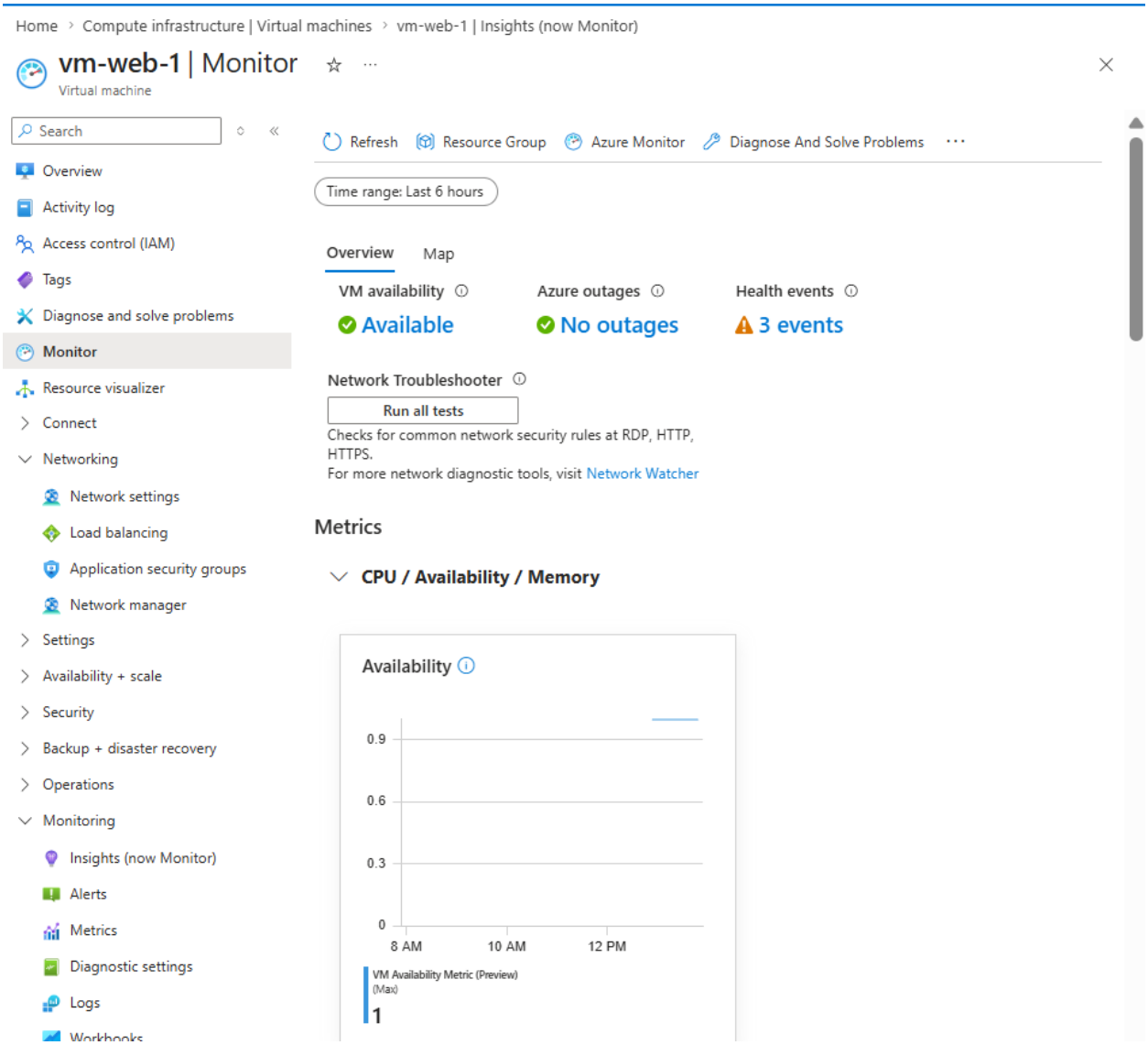
Create | Manage deleted vaults | Manage view | Refresh | Export to CSV | Open query | ... | Group by none

Filter for any field... | Subscription equals all | Resource Group equals all | Location equals all | Add filter

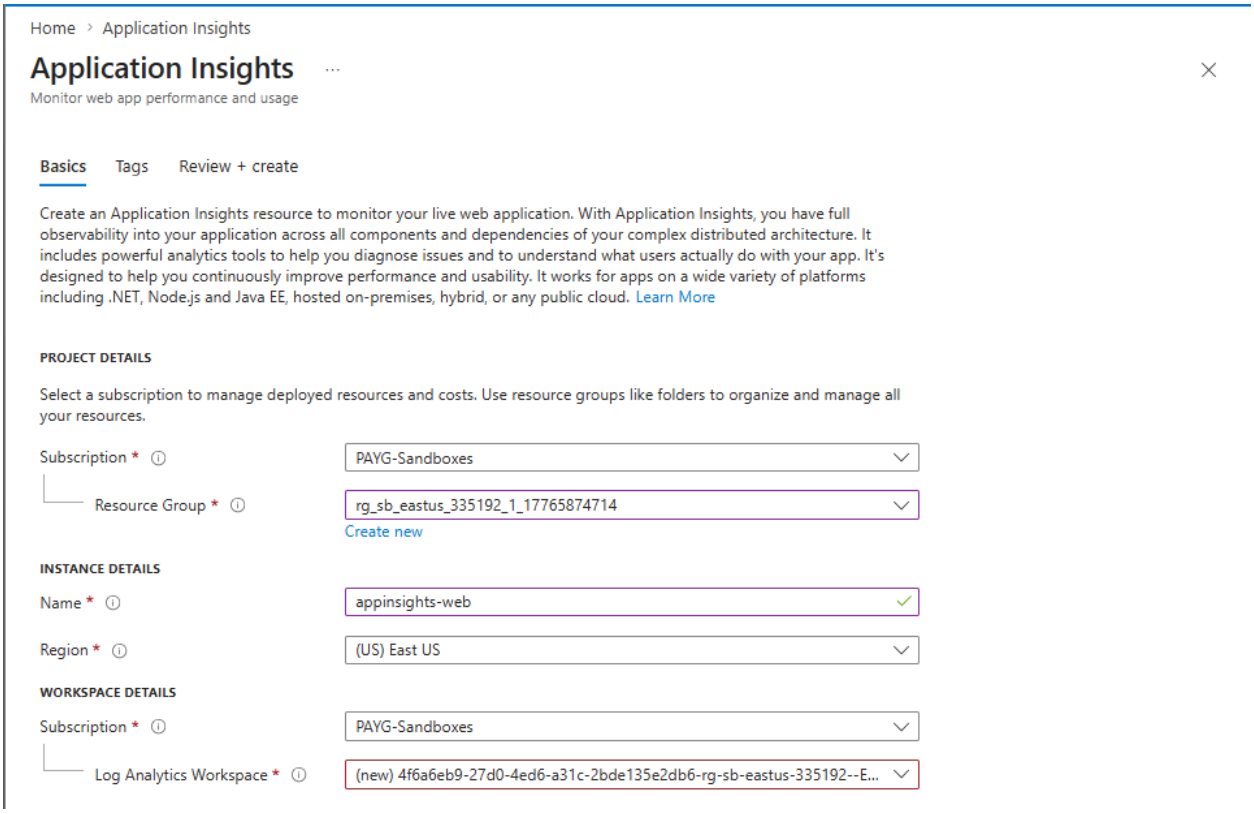
Name ↑	Type	Resource Group	Location	Subscription	Tags
kv-enterprise1104	Key vault	rg_sb_eastus_335...	East US	PAYG-Sandboxes	

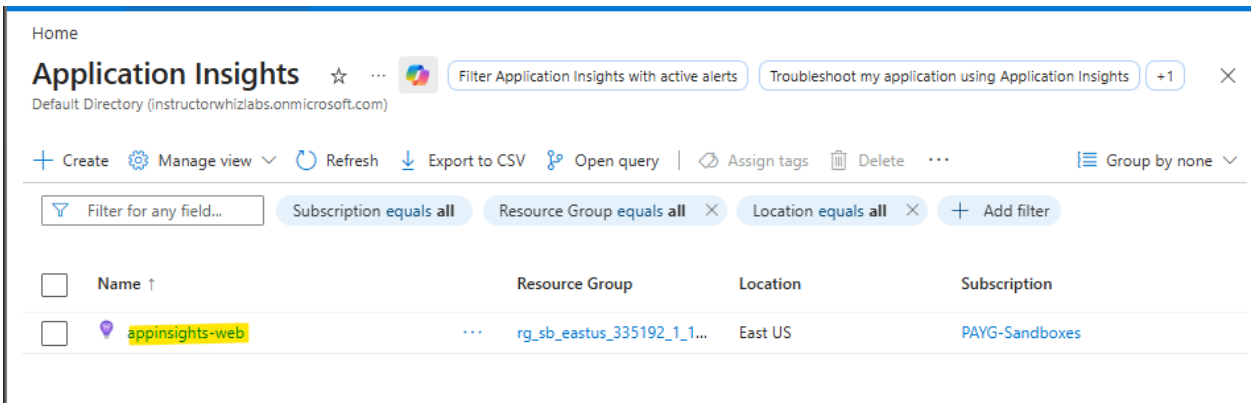
7. Monitoring & Logging

Enable Azure Monitor → monitoring is already enabled by default



Enable Application Insights





Configure at least one Alert

Home > Compute infrastructure | Virtual machines > vm-web-1 | Alerts

Create an alert rule

Scope | **Condition** | Actions | Details | Tags | Review + create

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Signal name * : Percentage CPU

See all signals

Alert logic

We have set the condition configuration automatically based on popular settings for this metric. Please review and make changes as needed.

Threshold type : Static / Dynamic

Aggregation type : Average

Value is : Greater than

Threshold * : 80 %

When to evaluate

Check every : 1 minute

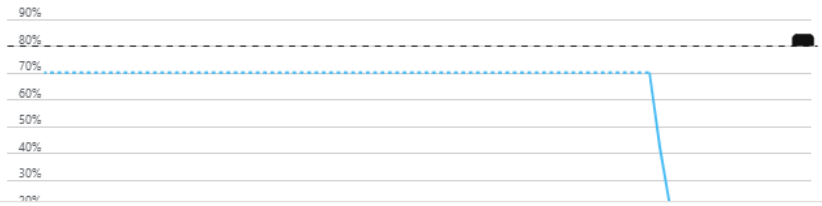
Lookback period : 5 minutes

Preview

$0.10 USD/month

Whenever the average Percentage CPU is greater than 80%

Preview time range : **Over the last 6 hours** | Time series : **Aggregate**



Review + create | Previous | Next: Actions >

Home › vm-web-1 | Alerts › Alert rules › high-cpu-alert

Edit alert rule

Scope | **Condition** | Actions | Details | Tags | Review + save

Configure when the alert rule should trigger by selecting a signal and defining its logic.

Signal name * : Percentage CPU

See all signals

Alert logic

Threshold type : Static / Dynamic

Aggregation type : Average

Value is : Greater than

Threshold * : 60 %

When to evaluate

Check every : 1 minute

Lookback period : 5 minutes

Preview $0.10 USD/month

Whenever the average Percentage CPU is greater than 60%

Preview time range : **Over the last 6 hours** | Time series : **Aggregate**

Home › Compute infrastructure | Virtual machines › vm-web-1

vm-web-1 | Alerts

Virtual machine

Which alerts require my immediate attention | Help me setup a new alert rule | +1

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
- Networking
  - Network settings
  - Load balancing

View as timeline (preview) | Create | Set up recommended alerts | Alert rules

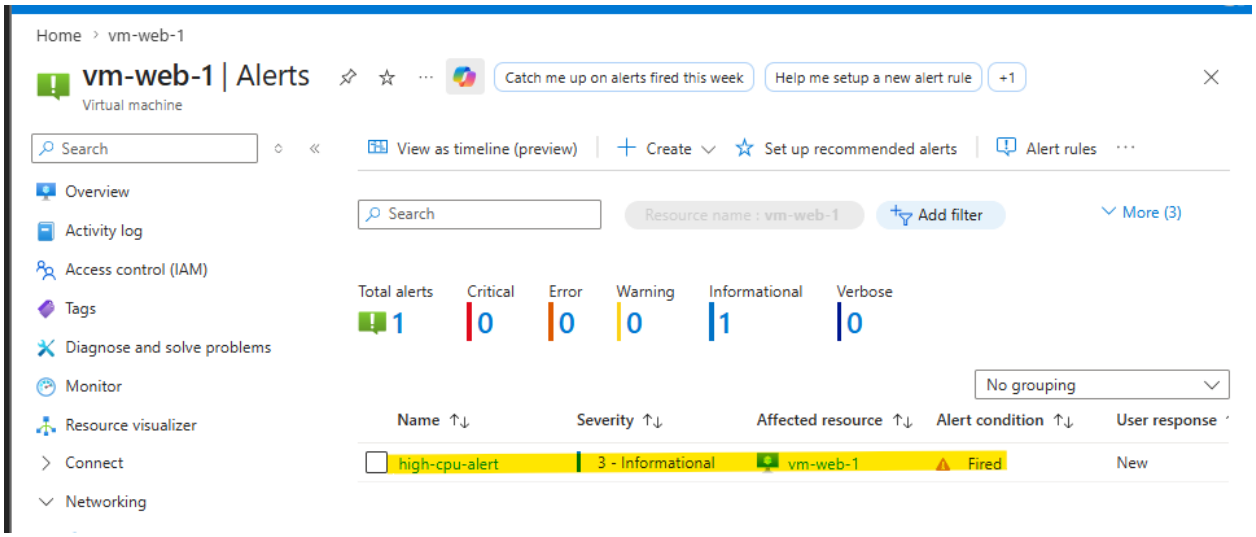
Search | Resource name : vm-web-1 | Add filter | More (3)

Total alerts	Critical	Error	Warning	Informational	Verbose
0	0	0	0	0	0

No grouping

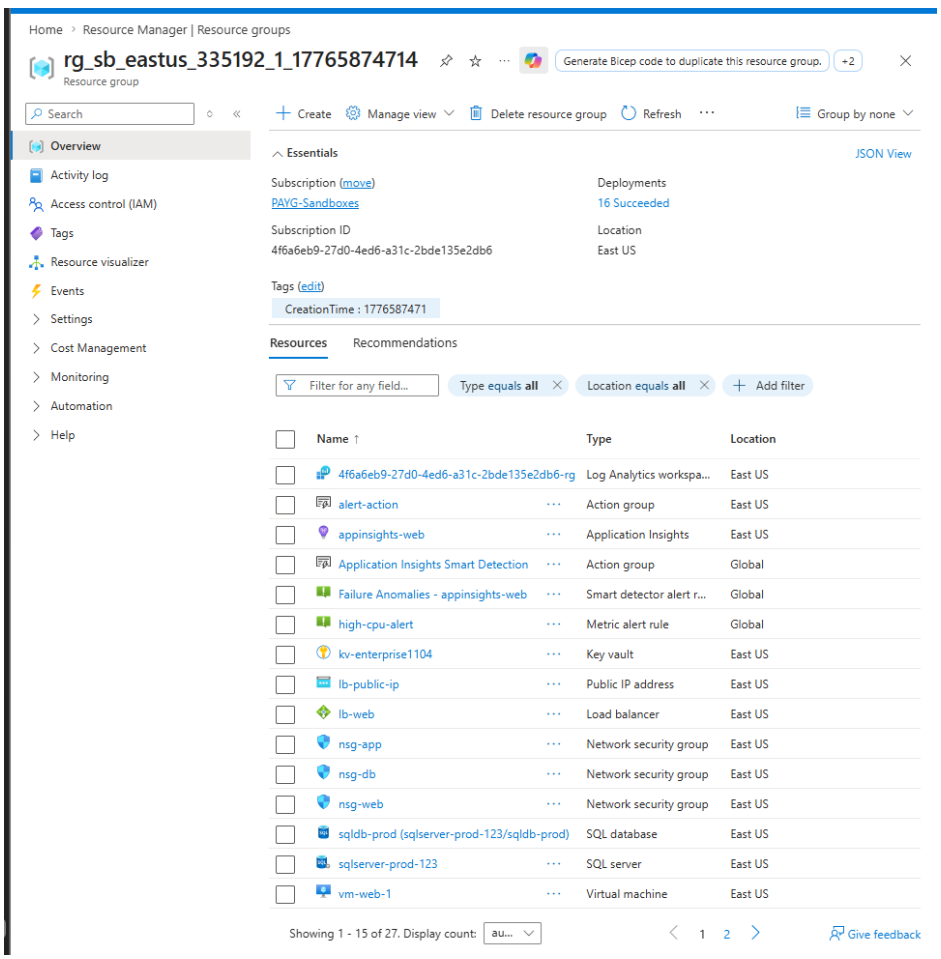
Name	Severity	Affected resource	Alert condition	User response

```
azureuser@vm-web-1: ~
yes > /dev/null &
yes > /dev/null &
[28] 2681
[29] 2682
[30] 2683
azureuser@vm-web-1:~$ yes > /dev/null &
yes > /dev/null &
yes > /dev/null &
[31] 2684
[32] 2685
[33] 2686
azureuser@vm-web-1:~$ yes > /dev/null &
yes > /dev/null &
yes > /dev/null &
[34] 2687
[35] 2688
[36] 2689
azureuser@vm-web-1:~$ yes > /dev/null &
yes > /dev/null &
yes > /dev/null &
[37] 2691
[38] 2692
[39] 2693
azureuser@vm-web-1:~$ yes > /dev/null &
yes > /dev/null &
yes > /dev/null &
[40] 2699
[41] 2700
[42] 2701
azureuser@vm-web-1:~$
```



8. Automation (Mandatory)

```
labuser_335192_1776587467903 [ ~ ]$ mkdir azure-capstone-tf1
labuser_335192_1776587467903 [ ~ ]$ cd azure-capstone-tf1
labuser_335192_1776587467903 [ ~/azure-capstone-tf1 ]$ nano main.tf
labuser_335192_1776587467903 [ ~/azure-capstone-tf1 ]$ terraform init
terraform apply
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/azurerm...
- Installing hashicorp/azurerm v4.69.0...
- Installed hashicorp/azurerm v4.69.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

No changes. Your infrastructure matches the configuration.

Terraform has compared your real infrastructure against your configuration and found no differences, so no changes
are needed.

Apply complete! Resources: 0 added, 0 changed, 0 destroyed.
labuser_335192_1776587467903 [ ~/azure-capstone-tf1 ]$
```



Home › Resource Manager | Resource groups › rg_sb_eastus_335192_1_17765874714

Resource Manager | Resource groups
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Search

- Resource Manager
- All resources
- Favorite resources
- Recent resources
- Resource groups
- Tags
- Organization
- Tools
- Deployments
- Help

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Name ↑
rg_sb_centralindia_335192_3_1776
rg_sb_eastus_335192_1_17765874
rg_sb_westus_335192_2_17765874

Showing 1 - 3 of 3. Display count: au...

rg_sb_eastus_335192_1_17765874714 | Export template | Template
Resource group

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- Overview
- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring
- Automation
  - Export template
- Help

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ExportTemplateCompletedWithErrors
Export template operation completed with errors. Some resources were not exported. Please see details for more information.

To export a selection of these resources, select resources from the Overview view and select the "Export template" option from the toolbar.

☑ Include parameters

Template Parameters

- Parameters (25)
- Variables (0)
- Resources (773)


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